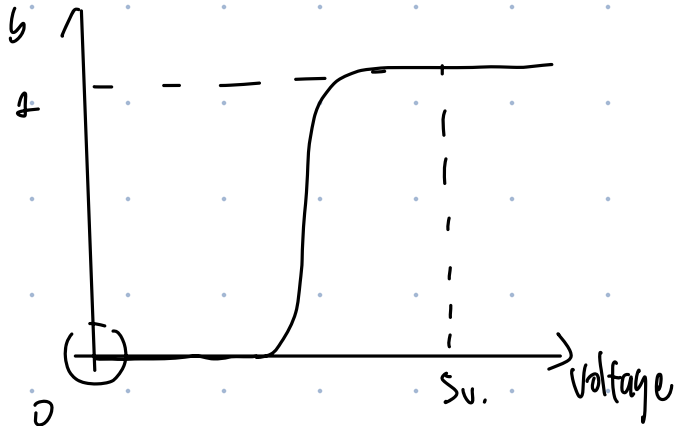


Lecture 24

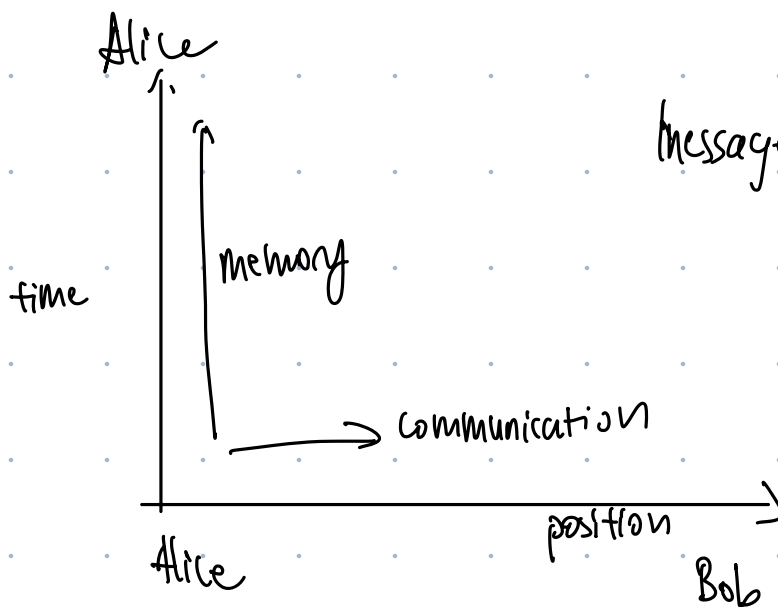
Quantum Error Correction

↳ Classical + Quantum Repetition Codes.

↳ No-cloning theorem



Noise 10^{-15}



message = message itself + redundancy

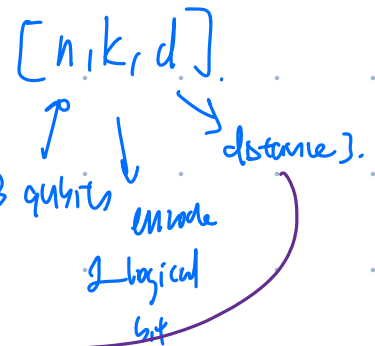
Simplest Code :

$\begin{cases} 0_L = 000 \\ 1_L = 111 \end{cases} \rightarrow \text{code rate: } 1/3.$

code distance = 2
 smallest # of errors
 that will take
 $0_L \rightarrow 1_L$

errors: $\begin{matrix} \text{sent} & \longrightarrow & \text{received} \\ 000 & \longrightarrow & 000 \\ 000 & \longrightarrow & 010 \\ 000 & \longrightarrow & 011 \\ 000 & \longrightarrow & 111 \end{matrix}$

$$\binom{4}{1} = \frac{4!}{2!1!}$$



errors that can be corrected

$$\left\lfloor \frac{d-1}{2} \right\rfloor$$

$$\frac{3-1}{2} = 1.$$

logical bit

Physical bits

Number of Errors during Transmission

BSC probability
 $0 \rightarrow 1, \epsilon$
 $1 \rightarrow 0, \epsilon$

Decoded logical bit using majority voting

0	000	0	$(1-\epsilon)^3$	$\underbrace{000}_{\text{receive}} \rightarrow \underbrace{000}_{\text{decode}}$
0	$\left\{ \begin{matrix} 001 \\ 010 \\ 100 \end{matrix} \right\}$	1	$3(1-\epsilon)^2\epsilon$	$\underbrace{001} \rightarrow \underbrace{000}$
0	$\left\{ \begin{matrix} 011 \\ 110 \\ 101 \end{matrix} \right\}$	2	$3\epsilon^2(1-\epsilon)$	$\underbrace{011} \rightarrow \underbrace{111}$
0	111	3	ϵ^3	$\underbrace{111} \rightarrow \underbrace{111}$

all these add up to 1!

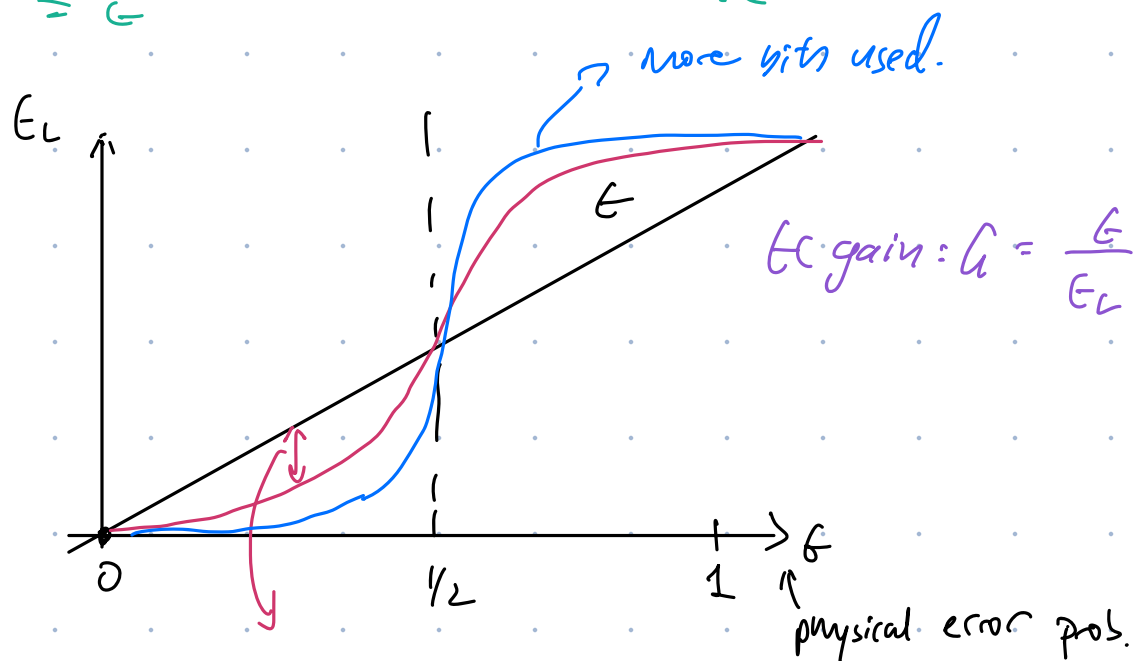
$$\begin{aligned}
 P(\text{failure}) &= 3\epsilon^2(1-\epsilon) + \epsilon^3 \\
 &= 3\epsilon^2 - 3\epsilon^3 + \epsilon^3 \\
 &= 3\epsilon^2 - 2\epsilon^3
 \end{aligned}$$

Without error code:

BSC failure prob
 $= \epsilon$

With error code

BSC failure prob
 $= 3\epsilon^2$



$$G \text{ gain: } G = \frac{\epsilon}{\epsilon_L}$$

$$\begin{aligned}
 G &\sim \frac{\epsilon}{3\epsilon^2} \\
 &\sim \frac{1}{3\epsilon}
 \end{aligned}$$

the lower the
 error rate, the
 better your gain.

$$[n, k, d] = [n, 1, n].$$

binomial distribution.

look $n \gg 1$,

mean # of errors for n -bits is $\underbrace{n\epsilon}$.

$\hookrightarrow$ ~~the~~ becomes gaussian distribution.

$$\epsilon = 10^{-2}, \quad n = 10^6.$$

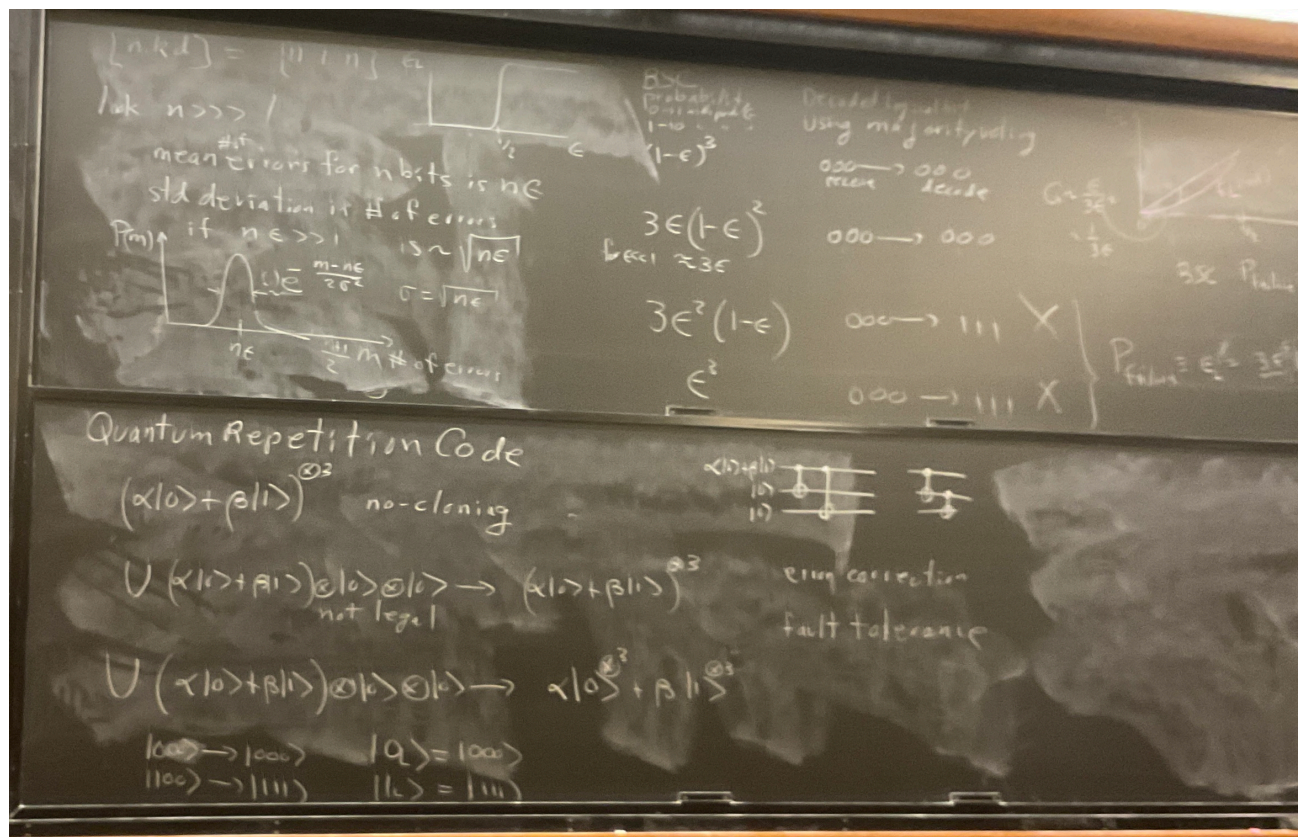
$$\text{avg. \# errors} = \epsilon n = 10^4.$$

$$\sigma \sim \sqrt{n\epsilon} = \sqrt{10^4} = 10^2.$$

Quantum Repetition Code

Fault tolerance (?)

↳ (



Logical Pauli Op

$$X_L = X \otimes X \otimes X$$

$$= |0_L\rangle\langle 1_L| + |1_L\rangle\langle 0_L|$$

$$Z_L = Z \otimes Z \otimes Z \text{ or } 1 \otimes 1 \otimes Z$$

$$\{X_L, Z_L\} = XXXZZZ + ZZZXXX = 0 \quad (?)$$

$$Y_L = iX_L Z_L$$